

Example of LLM for chemical planning

Prompt: You are given three different solutions (A, B, and C), each containing a different pair of inorganic salts from the following list, but you do not know which pair of salts are present in each solution: KCl, AgNO₃, Pb(NO₃)₂, BaI₂, Na₂SO₄, Ni(ClO₄)₂ in addition to the three unknown solutions, you have access to 1M KOH and 1M HNO₃ and no other solution. you can mix the unknown solutions together or add acid or base to them. Your task is to identify which pair of salts is present in each solution. Decompose this problem into a sequence of experiments. Explain which tests to perform, how to measure outcomes, and how to iteratively refine the hypotheses to reach the final conclusion.

Answer:

Step 1: Analyze tools and constraints we have

Observation types:

Solubility rules:

- Ag⁺: insoluble with Cl⁻ (AgCl ppt), soluble otherwise
- Ba²⁺: insoluble with SO₄²⁻ (BaSO₄ ppt)

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Step 2: Individual base test (add KOH)

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Identify Ag⁺ (brown/black), Pb²⁺ (white), Ni²⁺ (green) by ppt color

Step 3: Identify halides and sulfates via mixing

Mix unknown solutions in pairs (A+B, A+C, B+C).

Observe halide/sulfate precipitation:

AgCl (white ppt), PbI₂ (yellow ppt), PbSO₄ (white ppt), BaSO₄ (ppt).

Step 4: Dissolve ppts with HNO₃

Ag₂O, Pb(OH)₂, Ni(OH)₂ dissolve in HNO₃

- Take ppt formed in step 2
- If ppt dissolves: confirms the cation identity

Step 5: Iteratively refine assignment

Suppose after KOH addition:

A → green ppt → contains Ni²⁺

B → brown ppt → contains Ag⁺

.....

Then mix solutions:

B + A → white ppt → indicates Ag⁺ + Cl⁻ → solution B has Ag⁺, A has Cl⁻

.....

Combine observation:

A → Ni²⁺ + Cl⁻?

B → Ag⁺ + I⁻?

C → Pb²⁺ + SO₄²⁻?

Step 6: Cross-check final assignments

Mix assigned solutions → no new ppt unless expected.

Dissolve expected hydroxides with HNO₃ → confirms cation identity.

Double-check unusual cations like K⁺ and Na⁺ (remain in solution, soluble).

! **Incomplete domain knowledge:**
Omits Ag₂SO₄, AgI precipitation

! **Redundant step:**
Unnecessary confirmation steps with no new information

✗ **Chemical error:**
Ag⁺ and I⁻ cannot coexist in same solution (B),
Pb²⁺ and SO₄²⁻ cannot coexist in same solution (C)

! **Systematic incompleteness:**
Only partial ions are recognized, not 2 complete salts in one solution